

Sayan Dutta

BS-MS STUDENT, IIT PATNA · NLP RESEARCH · LLM REASONING · EMPATHETIC DIALOGUE SYSTEMS

Research Interests: LLM Reasoning · NLP · Controllable Generation · AI for Human-Centered Dialogue

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RESEARCH HIGHLIGHTS

- ACL ARR May 2026 Submission — Co-Author
- ECOT-ERG: Emotion-Cause CoT Framework (T5 + DPO)
- Research Intern, BITS Pilani — Dr. Srishti Gupta
- Chain-of-Thought Supervision for Empathetic Dialogue
- 16-Config Ablation · BERTScore · EASE · Cause-F1

EDUCATION

Indian Institute of Technology Patna

2025 – 2030

BS-MS, Computer Science and Data Analytics (Integrated 5-Year) · **Expected Graduation: 2030**

CGPA: 9.0 / 10.0 | **Coursework:** Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Optimization & Mathematical Foundations, Operating Systems, Database Systems

Research Experience: Machine Learning, Natural Language Processing, LLM Reasoning

RESEARCH OUTPUT

ECOT-ERG: Emotion-Cause Chain-of-Thought Framework for Empathetic Response Generation

Sayan Dutta, Srishti Gupta, and collaborators · **ACL ARR May 2026** (Under Review) · Supervisor: Dr. Srishti Gupta, PostDoctoral Fellow, CS, BITS Pilani

ROLE

Reasoning Pipeline Design

Emotion Prediction Module

Cause Identification Module

T5 Fine-Tuning

DPO Alignment

Ablation Studies

Code — GitHub

Paper — ARR

RESEARCH EXPERIENCE

Research Intern — BITS Pilani (Pilani Campus)

Apr 2026 – May 2026

Supervisor: Dr. Srishti Gupta, PostDoctoral Fellow, Dept. of Computer Science · Remote

ECOT-ERG: Emotion-Cause Chain-of-Thought Framework for Empathetic Response Generation

- Architected and implemented the ECOT-ERG reasoning framework, integrating emotion prediction, cause identification, chain-of-thought supervision, and T5-based response generation into a unified end-to-end pipeline.
- Built modular emotion prediction and cause identification components, enabling independent evaluation and ablation of each reasoning stage without retraining the full system.
- Fine-tuned T5 models with FP16 mixed-precision training; constructed DPO preference pairs from ranked model outputs for empathy alignment.
- Executed 16 experimental configurations (4-way ablation: CoT supervision × Emotion Prediction × Cause Identification × DPO); evaluated on BERTScore, EASE, Cause-F1, and Relevance-Accuracy.
- Co-authored ACL ARR May 2026 submission; conducted error analysis and generated qualitative outputs for paper discussion.

SELECTED PROJECTS

Blix.ai — Offline Persistent-Memory LLM Assistant

GitHub

- Built a fully air-gapped personalized LLM assistant running locally hosted Mistral-7B via Ollama — zero external API dependency.
- Investigated retrieval-aware memory ranking strategies for long-term personalized LLM interactions in offline settings.
- Prototyped embedding-based semantic retrieval using SentenceTransformers + FAISS for scalable long-term memory recall.
- Stack:** Python · Mistral-7B · Ollama · FAISS · SentenceTransformers · Express.js

ASES — AI-Driven Precision Agriculture Advisory Platform

GitHub

- Built crop and fertilizer recommendation engines using feature-normalized KNN over multi-dimensional soil and environment vectors.

- Integrated Google Gemini API with structured prompt engineering for contextual advisory generation; deployed via Streamlit + SQLite for lightweight local inference.
- **Stack:** Python · scikit-learn · Google Gemini API · Streamlit · SQLite

TECHNICAL SKILLS

FRAMEWORKS & LIBRARIES

PyTorch · HuggingFace Transformers · Seq2SeqTrainer · scikit-learn

LLM & REASONING

Chain-of-Thought Supervision · DPO · LoRA · RAG · Controllable Generation · Prompt Engineering

NLP

ERC · Empathetic Dialogue · Seq2Seq Learning · BERTScore · EASE · Cause-F1 · FAISS · SentenceTransformers

INFRASTRUCTURE

Linux · Git · CUDA · FP16/BF16 · W&B · Jupyter · Ollama · Google Colab

LANGUAGES

Python · C++

MATHEMATICS

Linear Algebra · Probability & Statistics · Optimization Fundamentals · Mathematical Reasoning

RESEARCH SKILLS

Experimental Design · Ablation Studies · Error Analysis · Statistical Evaluation · Scientific Writing · Reproducible Research Pipelines

RESEARCH INTERESTS

Natural Language Processing · LLM Reasoning & Controllable Generation · Emotion Recognition in Conversation (ERC) · Empathetic Dialogue Systems · Retrieval-Augmented Generation · Efficient Fine-Tuning (LoRA / DPO) · Optimization for AI Systems · Decision Making Under Uncertainty

Developing interest in: Multilingual NLP · Information Extraction · Question Answering · Low-Resource NLP

LEADERSHIP

Lead Google Student Ambassador, IIT Patna (GID: 986) · 2025–Present · Organized AI/ML workshops and developer events for 400+ students.